

Training V6-ROBUST Report: Fine-Tuning (Transfer Learning)

Overview

This report documents the results of Phase 3, Version 6 (V6-ROBUST). This experiment tested the **Transfer Learning** hypothesis.

Hypothesis: A large "Base Model" (V5, trained on 1,750 shot-type videos) can be used as a "Pro Cricketer" brain. By fine-tuning this brain on our small, 225-video "error" dataset, we can achieve high accuracy on error detection.

Result: The hypothesis was **proven incorrect**. The Transfer Learning model failed to learn the error classes, performing worse than our V4 model.

Dataset & Base Model Summary

- **Error Dataset:** 224 successfully loaded .npy feature files from our manually-labeled videos.
- **Base Model:** base_model_BRAIN_V5.keras (The "brain" from our 33.60% accuracy V5 Shot Classifier).
- **Scaler:** shot_scaler_V5.pkl (The scaler from the V5 model was used to keep data consistent).

Train/Validation/Test Split

A 3-way stratified split was used to ensure robust academic validation:

- **Training Samples:** 156
- **Validation Samples:** 34
- **Testing Samples:** 34

Model Training: Fine-Tuning (V6)

- **Algorithm:** Transfer Learning (Fine-Tuning).
- **Architecture:**
 1. Loaded the V5 "Base Brain" (base_model_BRAIN_V5.keras).
 2. Set base_brain.trainable = False. This "froze" all the knowledge from the 1,750-video dataset.
 3. Attached a new, trainable "Head" (two Dense layers) for our 5 error classes.

- **Training:**
 - The model was trained for 60 epochs with EarlyStopping (monitoring val_loss).
 - class_weight='balanced' was used to fight data imbalance.
 - The model trained quickly, but validation accuracy did not significantly improve.

Model Evaluation (V6)

Accuracy & F1 Scores

Metric	Test Set
Test Accuracy	29.41%

F1-Score (Weighted) 25.0%

Analysis of Failure

The overall accuracy (29.41%) is extremely low and misleading. The **Classification Report** tells the true story of a catastrophic failure.

Classification Report (Test Set)

Class	precision	recall	f1-score	support
good	0.50	0.58	0.54	12
head_falling	0.00	0.00	0.00	6
late_foot	0.00	0.00	0.00	4
low_elbow	0.00	0.00	0.00	4
off_balance	0.21	0.38	0.27	8
macro avg	0.14	0.19	0.16	34
weighted avg	0.23	0.29	0.25	34

Per-Class Notes & Conclusion

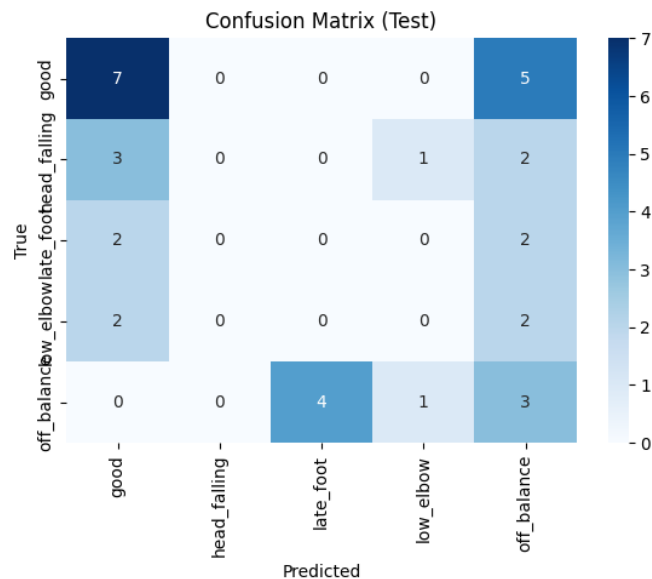
This is a complete model failure.

1. **Model is "Blind":** The model is **100% blind** to three of our five error classes (head_falling, late_foot, low_elbow), achieving an F1-score of **0.00** for all of them.

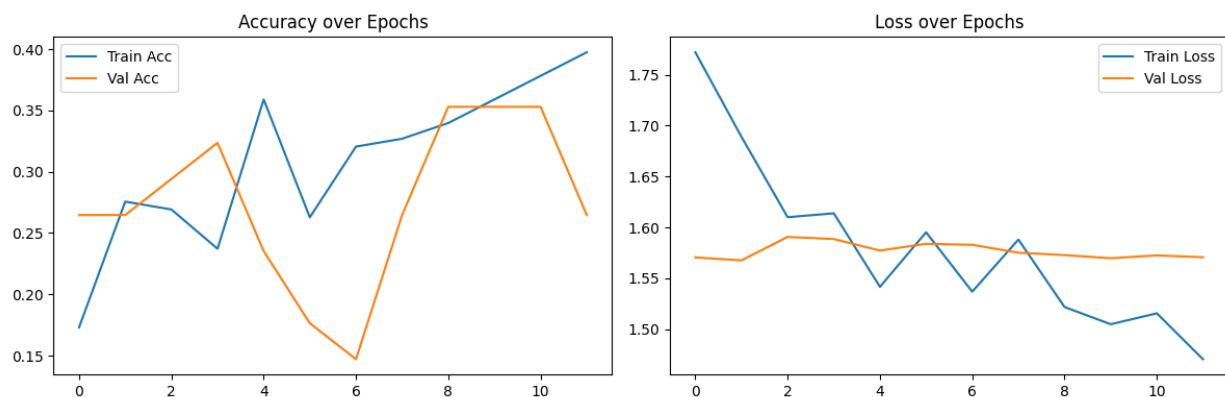
2. **It's Just Guessing:** The model only learned to distinguish (poorly) between good and off_balance. Its 29.41% accuracy is purely a result of these two guesses.
3. **"The Wrong Teacher" Hypothesis:** This failure proves our V5 "Base Brain" was the **wrong teacher**. It was trained to answer "What shot is this?" (e.g., "pull" vs. "sweep"). To do this, it learned to *ignore* fine-grained details like "head stability" or "foot timing" and focus only on the *broad direction* of the bat swing.
4. **V4 vs V6:** Our V4 model (trained from scratch on 225 videos) was *superior* to this V6 model, because it actually achieved F1-scores of 0.22 for late_foot and 0.20 for low_elbow.

Visualizations:

- **Confusion Matrix:**



- **Training History:**



Final Summary

Training V6 proves that **Transfer Learning is the incorrect hypothesis** for this problem. The "Pro Cricketer" brain (V5) does not contain the necessary knowledge to be a "Coach" (V6).

This confirms that our V4 model (LSTM trained from scratch) was the **correct architecture**, and its only problem was "**Data Starvation.**"

Recommended Next Steps

We must abandon the Transfer Learning approach. We will return to the V4 architecture and solve its "Data Starvation" problem by **manufacturing more data**.

1. **Data Augmentation:** We will create a new script to "augment" our 225 .npy files (using flips, jitter, and time-shifts) to create a new, larger dataset of ~1,000+ samples.
2. **SMOTE:** We will apply SMOTE to this new, *larger* dataset to balance it.
3. **Final Model (V7):** We will train the V4 LSTM architecture on this new, large, augmented, and balanced dataset.